

Erdős Problem #346

EP-346

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $A = \{1 \leq a_1 < a_2 < \dots\}$ be a set of integers such that

{UL}

{LI} $A \setminus B$ is complete for any finite subset B and {/LI}

{LI} $A \setminus B$ is not complete for any infinite subset B .{/LI}

{/UL}

(Here 'complete' means all sufficiently large integers can be written as a sum of distinct members of the sequence.)

Is it true that if $a_{n+1}/a_n \geq 1 + \epsilon$ for some $\epsilon > 0$ and all n then $\lim_n \frac{a_{n+1}}{a_n} = \frac{1+\sqrt{5}}{2}$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>